Energetic and Frictional Effects in the Transport of Ions in a Cyclic Peptide Nanotube

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The effects of geometric restraints and frictional parameters on the energetics and dynamics of ion transport through a synthetic ion channel are investigated using molecular dynamics (MD) simulations for several different ions. To do so, potential of mean force profiles and position-dependent diffusion coefficients for Na^+, K^+, Ca^{2+}, and Cl^- transport through a simple cyclic peptide nanotube, which is composed of 4× cyclo[–(D-Ala-Glu-D-Ala-Gln)$_2$–] rings, are calculated via an adaptive biasing force MD simulation method and a Baysian inference/Monte Carlo algorithm. Among the restraints and parameters examined in this work, the radius parameter used in the flat-bottom half-harmonic restraint at the entrance and exit to channel has a great effect on the energetics of ion transport through the variation of entropy in the outside of the channel. The diffusivity profiles for the ions show a strong dependence on the damping coefficient, but the dependence on the coefficient becomes minimal inside the channel, indicating that the most important factor which affects the diffusivity of ions inside the channel is local interactions of ions with the structured channel water molecules through confinement.

Keywords: cyclic peptide nanotube, ion channel, molecular dynamics simulation, potential of mean force (PMF), position-dependent diffusion coefficient

Introduction

Ion transport through ion channels in cell membranes is a crucial biological process that is involved in several physiological phenomena such as nerve and muscle excitation and signal transduction. Due to the importance of this process, a great number of experimental and computational studies have been devoted to understanding its underlying mechanism.[1]

From a theoretical and computational point of view, ion transport through ion channels can be analyzed on a basis of the free energy variation and diffusivity of ions occurring during transport.[2–6] For example, the free energy change and local diffusivity of ions through ion channels enable us to calculate ion permeability and conductance, providing insight into the selectivity of ion channels to a specific ion.[4,7–9] In this respect, it is essential to obtain accurate profiles for the free energy change and local diffusivity of ions through ion channels in order to clearly understand the mechanism of the process.

Molecular dynamics (MD) simulations are a powerful tool with which to explore complex systems such as biological systems at atomistic levels, and have been popularly used to assess the energetics and dynamics of ion transport through natural and synthetic ion channels.[4,10–17] Allen and coworkers examined the transport of K^+ through Gramicidin A channels by calculating the potential of mean force (PMF) for the ion via MD simulations.[4] Hwang *et al*., Dehez *et al*., and Yan *et al*. independently employed MD simulations to obtain the energetics of ion transport through self-assembling α-cyclic peptide nanotubes (CPNs), which is one type of synthetic ion channels.[14,15,17] Although MD simulations provide detailed information on processes occurring in biological systems, the consequences from MD simulations depend deeply on simulation input parameters and restraints applied in the simulations.[18]

In this work, we investigate the effects of geometric and dynamical restraints in MD simulations as well as ion charges and valences on the energetics and diffusivity of ion transport through a simple CPN in water.[19–21] We choose CPNs as a target system because they have simple tubular structures like natural ion channels, thus providing a good chance to examine the energetics and diffusivity of ion transport through ion channels at reasonable computational cost. We calculate the PMFs and position-dependent diffusion coefficient profiles for Na^+, K^+, Ca^{2+}, and Cl^- through a simple CPN solvated in water using an adaptive biasing force (ABF) MD simulation method and a Baysian inference/Monte Carlo (BI/MC) technique with a different set of simulation parameters and restraints.[22,23] Although CPNs allow only cations to pass through themselves,[24,25] Cl^- was included in our calculations to look into the effect of ion valence on the energetics and diffusivity of the ion. In addition, the relationship between the energetics and dynamics of ions is discussed on a basis of the obtained PMF and position-dependent diffusion coefficient profiles.

The paper is organized as follows. In the next section, the simulation systems are introduced and computational

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methods for the construction of PMF profiles and position-dependent diffusion coefficients for Na^+, K^+, Ca^{2+}, and Cl^- are briefly explained. The results are analyzed and discussed in the Results and Discussion section and conclusions are made in the final section.

Computational Methods

Since the details of the simulation preparation are described in Ref. 26, we will only briefly address the system and procedure herein. The model system includes one CPN, 1130 water molecules, and a single ion such as Na^+, K^+, Ca^{2+}, or Cl^- in a simulation box with 31.5 Å × 31.5 Å × 38.0 Å dimensions. The CPN is formed by 4× cyclo[–(D-Ala-Glu-D-Ala-Gln)$_2$–] rings through hydrogen bonds shown in Figure 1. The protonated form of glutamic acid is used in the CPN. The channel axis of the CPN was aligned parallel to the z-axis of the system. The number of water molecules in the system was calculated assuming a partial molar volume of 1.00 cm^3/g for water and 0.73 cm^3/g for the CPN.[27] All the simulations were performed using the canonical ensemble (NVT ensemble) and periodic boundary conditions in all directions. The system was fully equilibrated at 298 K using a Langevin thermostat with a damping coefficient, γ_L, of 4.0 ps^{-1}. We chose γ_L = 4.0 ps^{-1} because the ABF trajectories of ions with that value of γ_L give rise to diffusion coefficients of ions in bulk water very close to the experimental results (Table 1).[28] To assess the effect of the damping coefficient on the diffusion coefficients of ions, MD simulations were also carried out with γ_L = 1.0 ps^{-1}. The CHARMM27 force field parameter sets for the CPN and the TIP3P water model for water were used,[29,30] and the force field parameters for ions were obtained from Beglov and Roux.[31] All the energy minimizations and MD simulations were performed using the NAMD 2.10,[32] and visualizations were made possible using the VMD 1.9.0.[33]

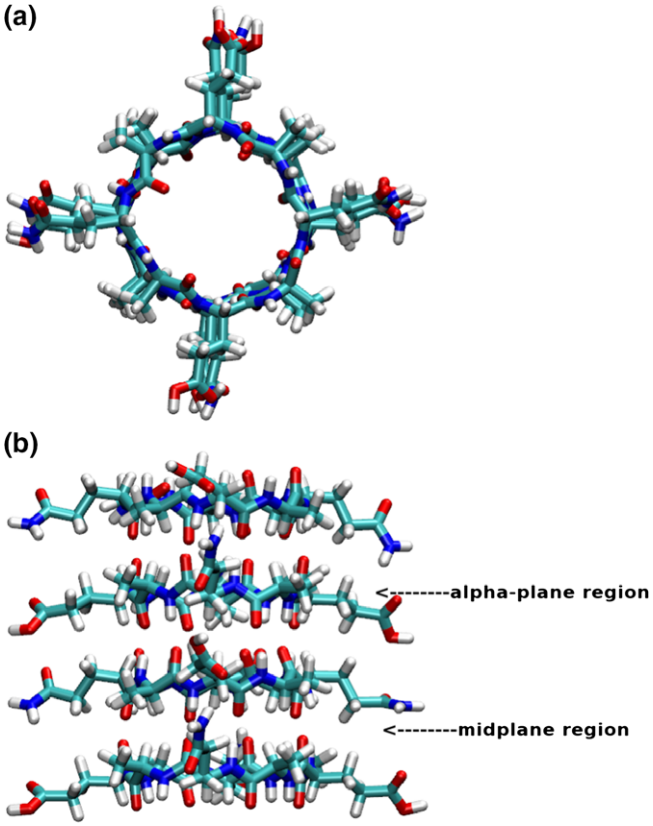


Figure 1. (a) Top view and (b) side view of a CPN, 4× cyclo [–(D-Ala-Glu-D-Ala-Gln)$_2$–]. The α-plane regions and midplane regions are defined in the text following the work in Ref. 27.

Table 1. Diffusion coefficients of ions in bulk water.

(Å^2/ps)	Na^+	K^+	Ca^{2+}	Cl^-
D_{abf} (4 ps)	0.141	0.203	0.0906	0.203
D_{abf} (1 ps)	0.187	0.274	0.132	0.275
D_{exp}[a]	0.133	0.196	0.0792	0.203

[a] Taken from Ref. 28.

PMF profiles of ions, $\Delta A(z)$, along the channel axis were calculated using the ABF method. A colvar method was used to maintain the channel axis parallel to the z direction. The distance between the two adjacent cyclic peptide rings was kept 4.7 Å in the z direction by restraining the center of mass of the alpha carbons in each ring at −7.05, −2.35, 2.35, and 7.05 Å with a harmonic spring constant of k_z^{ring} = 25.0 kcal/mol/Å^2. Another spring constant of k_z^{ring} = 1.0 kcal/mol/Å^2 was also used to investigate the effect of the channel rigidity of the CPN on the PMF and position-dependent diffusion coefficient profiles. To circumvent the sampling problem due to the integration over all xy extents and to keep the ions within a certain area outside the channel,[4,5] a flat-bottom half-harmonic restraint was applied to the ions in the xy-plane as shown below,

$$V_{fb}(x,y) = \begin{cases} 0, & \text{if } x^2 + y^2 \le R_0^2 \\ \frac{1}{2}k_{xy}^{ion}\left(\sqrt{x^2+y^2} - R_0\right)^2, & \text{otherwise} \end{cases} \quad (1)$$

where k_{xy}^{ion} = 10 kcal/mol/Å^2 and R_0 = 2.5 Å for all the ions.[34] To examine the dependence of the PMF and position-dependent diffusion coefficient profiles on R_0, we also performed ABF simulations with R_0 = 4.5 Å for Na^+, K^+, and Ca^{2+}. In the case of Cl^-, a smaller radius of R_0 = 3.5 Å was used because Cl^- failed to enter the channel and was found outside the CPN when R_0 = 4.5 Å was applied. Instead of the flat-bottom half-harmonic restraint, a conventional harmonic restraint is sometimes applied to ions in xy-plane.[14,26] To compare the flat-bottom half-harmonic restraint with the conventional harmonic restraint method, we also used a harmonic restraint with a spring constant, k_{xy}^{ion} of 0.16 kcal/mol/Å^2 to K^+ and Na^+ in xy-plane to keep them within an average distance of 2.5 Å from the channel axis when the ions move out of the channel.

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The BI method was used to compute position-dependent diffusion coefficients of Na^+, K^+, Ca^{2+}, and Cl^- along the channel axis (z axis) $D(z)$.[9,22] In the method it is assumed that the evolution of an ion, which undergoes a free diffusional motion in an ABF MD simulation, can be completely described by PMF and position-dependent diffusion coefficient profiles for that ion. Then a best set of unknown parameters are searched via a MC technique to give rise to the likelihood function and prior probability with highest values, and the final set of the unknown parameters are used to construct the PMF and diffusivity profiles. For further details, refer to Ref. 22.

In our MD simulations, the reaction pathway of each ion spanned from −22.0 to 22.0 Å and was divided by 111 bins with a bin width of 0.4 Å. For a PMF profile and a position-dependent diffusion coefficient for each ion under a specific condition, four independent simulations (two independent simulations in case of Cl^-) were carried out and in each simulation no fewer than 500 000 samples were collected per bin. The final symmetrized PMF profiles shown in the next section were constructed by creating duplicate bins on opposite sides of the channel. A time step of $\Delta t = 1.2$ ps was chosen and the functions $D(z)$ and $\Delta A(z)$ were discretized and interpolated by piecewise cubic polynomials with a bin width of 0.4 Å from −22.0 to 22.0 Å.

Results and Discussion

Figure 2 depicts the symmetrized PMF profiles for a single Na^+, K^+, Ca^{2+}, and Cl^- through the model CPN that have been calculated using the ABF, TI, and BI/MC methods, respectively. For comparison, the PMF profiles for Na^+, K^+, and Cl^- that were previously obtained using the umbrella sampling (US) method are also shown.[26] To acquire the PMF curves from the ABF simulations, a damping coefficient of $\gamma_L = 4.0$ ps^{-1} was used in the Langevin thermostat and a harmonic restraint of $k_z^{ring} = 25$ kcal/mol/Å^2 was applied to each cyclic peptide ring in the z direction. In addition, a flat-bottom half-harmonic restraint of $R_0 = 2.5$ Å was used to keep ions within 2.5 Å from the channel axis in the xy-plane. The PMF curves calculated from each method show good agreement with those from the other methods. As already noted, the PMF profiles for Na^+, K^+, and Cl^- indicate strong selectivity of the CPN to Na^+ and K^+ transport over Cl^-.[26] Although there exist negative free energy minima in the PMF profiles for Ca^{2+}, those minima are not as deep as those for Na^+ or K^+, and the barriers are higher, so transport of the divalent ion will be suppressed. Embedding the CPN in a lipid bilayer is unlikely to change this result.[17] The marked free energy minima at the second and third α-plane regions in the PMF profiles for Ca^{2+} imply that Ca^{2+} is stabilized by surrounding water molecules located at the first and second midplanes because more water molecules can exist in midplanes than in α-planes and the divalent Ca^{2+} favors a higher hydration energy than monovalent Na^+ or K^+.

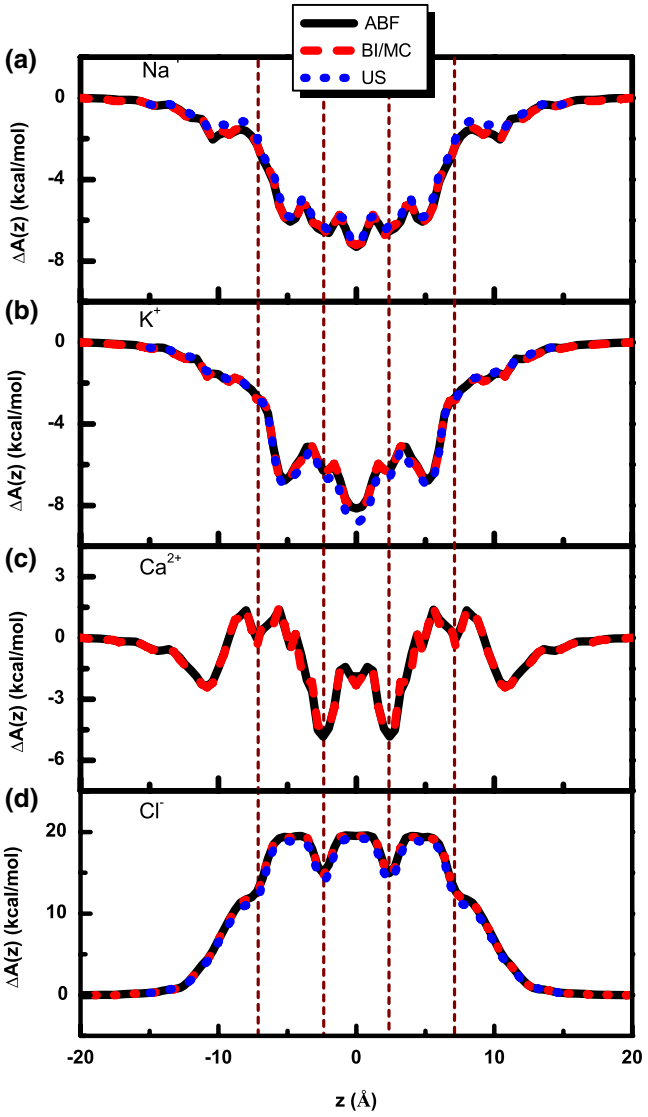


Figure 2. Final symmetrized PMF profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- obtained from the ABF and BI/MC methods, respectively. For Na^+, K^+, and Cl^-, comparisons with the US results in Ref. 26 are also shown. For parameters, $\gamma_L = 4.0$ ps^{-1}, $k_z^{ring} = 25.0$ kcal/mol/Å^2, and $R_0 = 2.5$ Å were used. See the text for details. The vertical broken lines indicate the position of each cyclic peptide ring in the z direction.

The dependence of the PMF profiles on the damping coefficient, γ_L, that is used in the Langevin thermostat is illustrated in Figure 3. Here we used two different values of γ_L, namely, $\gamma_L = 4$ ps^{-1} and $\gamma_L = 1$ ps^{-1}. The other restraints such as k_z^{ring} and R_0 are the same as in Figure 2. Figure 3 demonstrates that the PMF profiles for the ions are insensitive to the change in γ_L. It can be speculated that the damping coefficient in the Langevin thermostat introduces friction to the constituents in the system and this affects dynamic properties such as diffusion coefficients of the ions, but it has little effect on static properties like PMF profiles.

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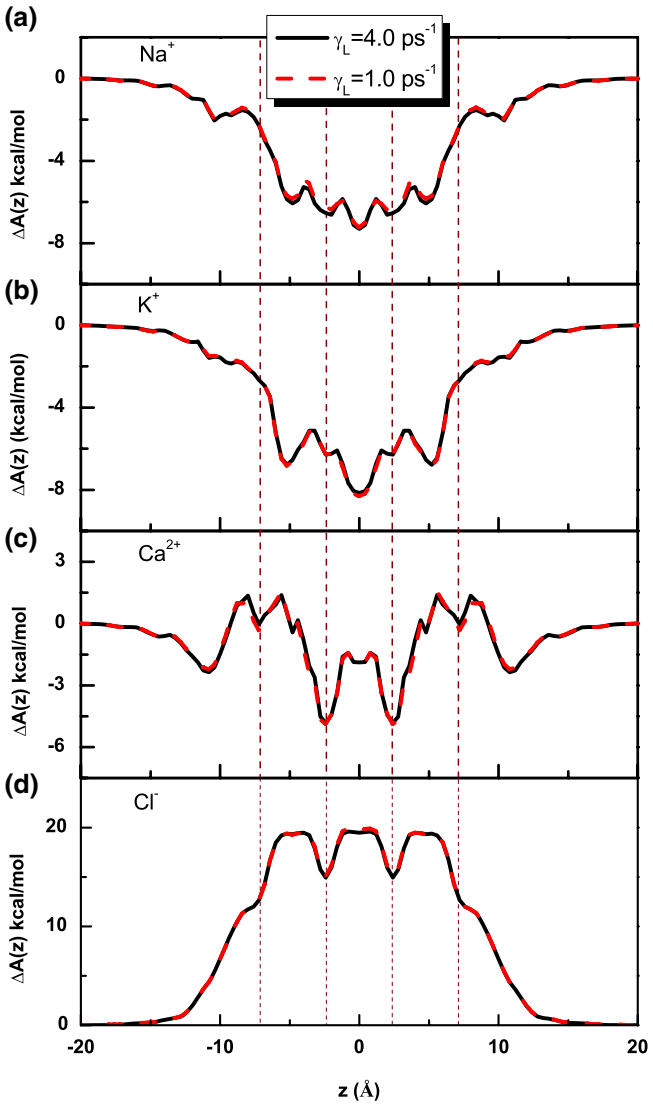


Figure 3. PMF profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- showing the dependence of PMF profiles on the damping coefficient (γ_L) in the Langevin thermostat method. For other parameters, $k_z^{ring} = 25.0$ kcal/mol/Å^2 and $R_0 = 2.5$ Å were used. The notation is the same as in Figure 2.

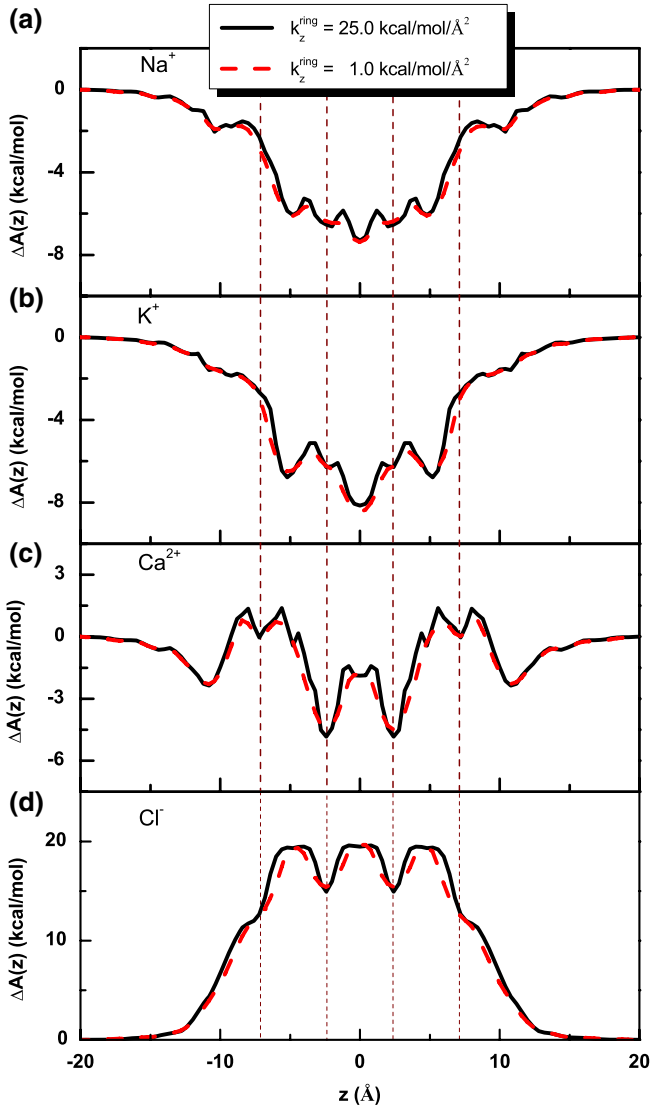


Figure 4. PMF profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- representing the dependence of PMF profiles on channel rigidity. The channel rigidity was determined by applying harmonic restraints with a spring constant of k_z^{ring} to the alpha carbons in the backbone of the CPN. For other parameters, $\gamma_L = 4.0$ ps^{-1} and $R_0 = 2.5$ Å were used. The notation is the same as in Figure 2.

The influence of the rigidity of CPN backbones on the PMF curves is presented in Figure 4. To do so, harmonic restraints with a spring constant, k_z^{ring} of either 25.0 kcal/mol/Å^2 or 1.0 kcal/mol/Å^2 were applied to the alpha carbons of each cyclic peptide ring in the z direction with the other restraints fixed as in Figure 2. When a weaker harmonic restraint of $k_z^{ring} = 1.0$ kcal/mol/Å^2 is applied, the local energy barriers and energy wells in the PMF profiles for all the cations tend to decrease slightly, making the overall shape of the PMF curves smoother. This indicates that as the rigidity of the CPNs is reduced, the cations are less confined and can move more freely inside the channel. In the case of Cl^-, a weaker harmonic restraint hardly changes the local minima and maxima in the PMF profile, but makes the free energy of Cl^- much lower in the other regions inside the channel. This may be related to the fact that Cl^- whose size is bigger than the other cations can make more room and become more stable inside the channel as a weaker harmonic restraint is applied.

As mentioned before, one dimensional PMF profiles of ions across ion channels are not well defined outside the channel due to the difficulty in sampling over the entire xy-plane. To circumvent the sampling problem, a flat-bottom half-harmonic restraint, which restricts the movement of ions within a proper area in the xy-plane was applied to the target ion. The effect of R_0 in Eq. (1) on the PMF profiles is investigated and shown in Figure 5. It is demonstrated that the minimum of the profiles rises as R_0 increases. When a larger value of R_0 is used, the sampling space for

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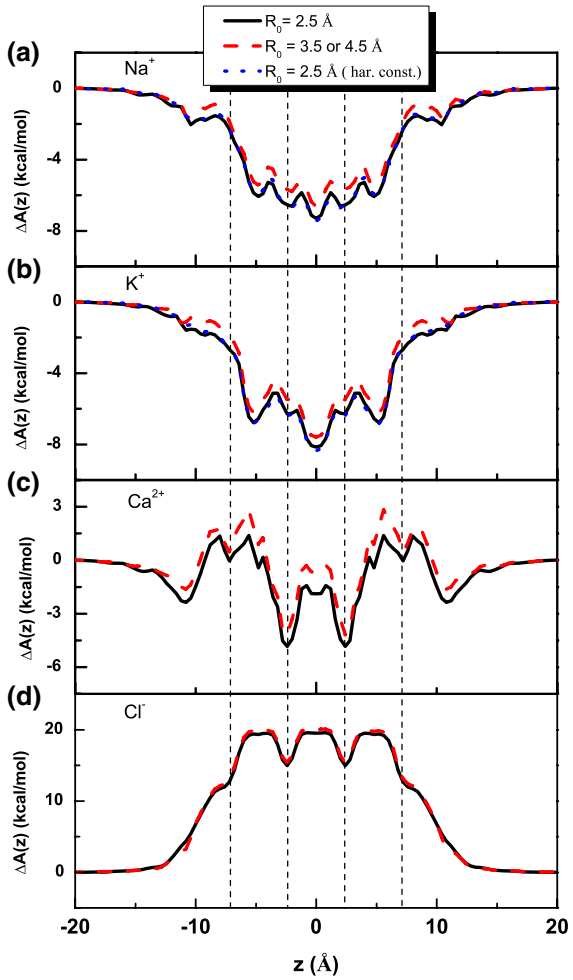


Figure 5. PMF profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- representing the dependence of PMF profiles on R_0 used in the flat-bottom half-harmonic restraints applied to the ions in the *xy*-plane. R_0 = 2.5 Å was used for all the ions, and R_0 = 4.5 Å was used for Na^+, K^+, and Ca^{2+}. In the case of Cl^-, R_0 = 3.5 Å was used instead. For other parameters, γ_L = 4.0 ps^{-1} and k_z^{ring} = 25.0 kcal/mol/Å^2 were used. The notation is the same as in Figure 2.

the ion is increased outside the channel, and this leads to the increase of the configurational entropy and the subsequent decrease of the free energy outside the channel. That restraint, however, has little effect on the configurational entropy and the free energy of ions inside the channel because the sampling space remains unchanged due to the confinement of ions inside the channel irrespective of the restraint. As a result, the free energy difference between inside and outside the channel is reduced as a larger R_0 is used. We can estimate the extent of the change of the free energy, ΔΔA, due to the increase of R_0 in the following way. Assuming that the internal energy of the cations remains unaltered as R_0 increases from 2.5 to 4.5 Å, the only contribution to ΔΔA stems from the increase of the configurational entropy, ΔΔS. Thus, $\Delta\Delta A \approx |-T\Delta\Delta S| = (298K)(k_B)\ln[(4.5Å)^2/(2.5Å)^2] \approx 0.70$ kcal/mol where k_B is the Boltzmann constant. This estimated value agrees well with the increases of the free energy minimum in the PMF profiles for the cations as R_0 increases in Figure 5.

Position-dependent diffusion coefficients for Na^+, K^+, Ca^{2+}, and Cl^- transport through the simple CPN and their dependence on γ_L in the Langevin thermostat are represented in Figure 6. To calculate the position-dependent diffusion coefficients of the ions, the BI/MC algorithm combined with the ABF simulation trajectories were used.[22] It is observed that the decrease of the position-dependent diffusion coefficients for all the ions begins at more than 10 Å away from the channel entrance and continues until the coefficients take their minimum values at the channel entrance regardless of the value of γ_L.[35] Although the diffusion coefficients of the ions in bulk water represent a notable dependence on γ_L, the dependence seems to be considerably diminished inside the channel. This indicates that interactions of ions with the backbone of the CPN and with channel waters inside the channel are more dominant, leading to reduction of the diffusion coefficients, and making the role of γ_L less effective inside the channel. Figure 6 also reveals that despite a distinct difference in the PMF profiles between K^+ and Cl^-, the diffusion coefficient profiles for both ions are much alike. This implies that unlike the PMF profiles which are generally governed by interactions of ions with global environments, the position-dependent diffusion coefficients of ions are more affected by local interactions of ions with water through confinement.

One issue in the position-dependent diffusion coefficient of an ion is whether the relative diffusivity, which is a position-dependent diffusion coefficient scaled by a calculated bulk value, remains constant regardless of γ_L in the Langevin thermostat.[5] If the relative diffusivity shows little dependence on γ_L, any position-dependent diffusion coefficient based on a specific value of γ_L can be estimated using the relative diffusivity profile and the bulk diffusion coefficient of the ion obtained with the same γ_L. In order to investigate the dependence of the relative diffusivity on γ_L, the calculated relative diffusivities for Na^+, K^+, Ca^{2+}, and Cl^- are illustrated in Figure 7. The relative diffusivities are also plotted using a fitting function form given as:

$$f(z) = \frac{1}{N}\left(a_2 + \frac{a_1 - a_2}{1.0 + e^{-(|z|-z_0)/dz}}\right), \quad (2)$$

where a_1, a_2, z_0, and dz are fitting parameters. As shown in Figure 7, the relative diffusivity profiles for all the four ions show a noticeable dependence on γ_L, which indicates that the relative diffusivity is not a universal function, but depends on simulation parameters like γ_L.

The effect of channel rigidity on the position-dependent diffusion coefficients of ions is illustrated in Figure 8. Here the effect of the channel rigidity was implemented by applying harmonic restraints with two different k_z^{ring} to the alpha carbons in the backbone of the simple CPN in *z* direction. As shown in Figure 8, the channel rigidity has a negligible

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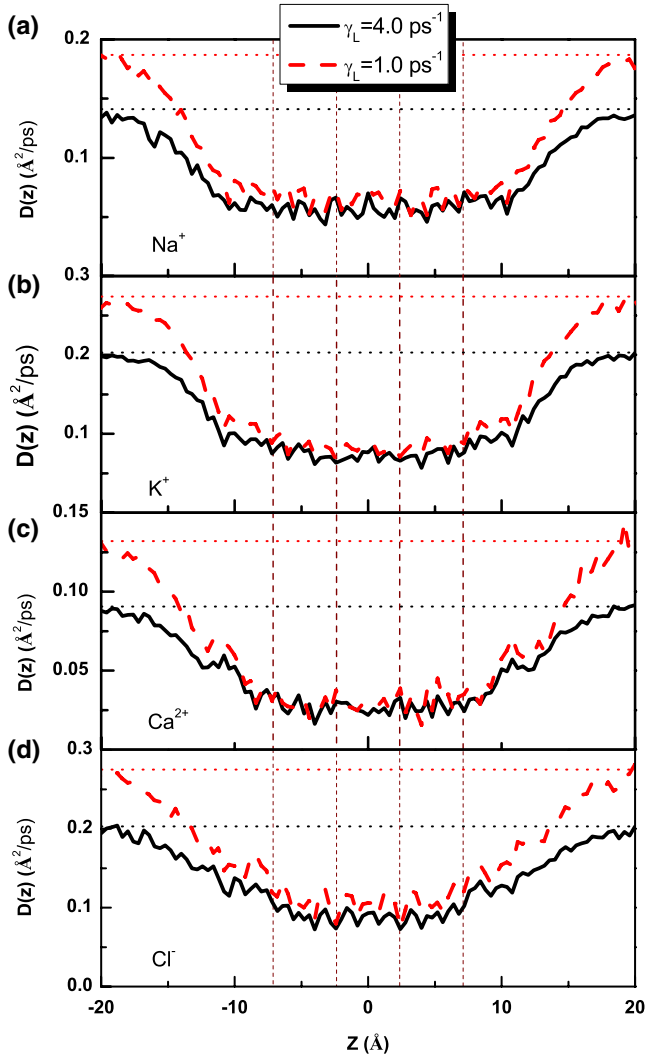


Figure 6. Position-dependent diffusion coefficient profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^{-1} as a function of axial direction z showing the dependence of diffusion coefficients on the damping coefficient (γ_L) in the Langevin thermostat. For other parameters, $k_z^{ring} = 25.0$ kcal/mol/Å^2 and $R_0 = 2.5$ Å were used. The horizontal broken lines indicate the diffusion coefficients of the ions in bulk water that were obtained using the ABF simulations followed by the BI/MC algorithm with each γ_L. The vertical broken lines are the same as in Figure 2.

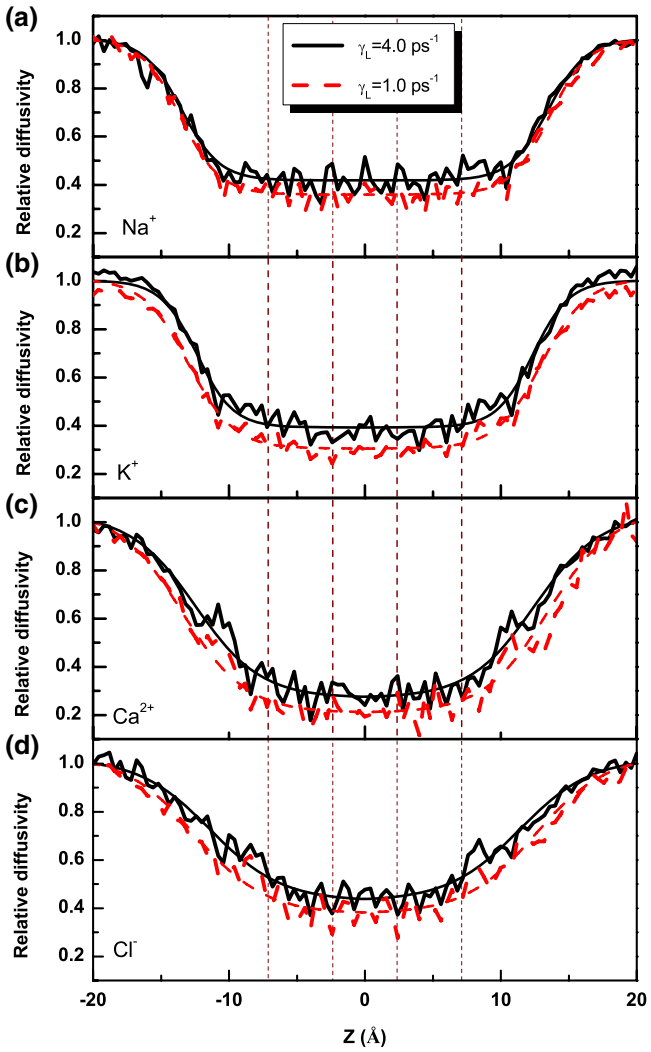


Figure 7. Relative position-dependent diffusion coefficient profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^{-1} as a function of axial direction z showing the dependence of relative diffusion coefficients on γ_L. The smooth lines are fits to the relative diffusivity functions that were plotted with Eq. (2). The vertical broken lines are the same as in Figure 2.

influence on the position-dependent diffusion coefficients of Na^+, K^+, and Ca^{2+}, but shows a slight effect in the case of Cl^-. These results and Figure 4 indicate that the channel rigidity of the simple CPN can change the energy landscapes of the cations to a moderate extent, but has a negligible influence on the dynamics of the cations. In the case of Cl^-, a weaker restraint of $k_z^{ring} = 1.0$ kcal/mol/Å^2, which has lowered the free energy of Cl^- considerably, leads to the slowdown in diffusivity of Cl^- inside the channel.

Finally, the dependence of the position-dependent diffusion coefficients of the ions on R_0 that is used in the flat-bottom half-harmonic restraint in Eq. (1) is displayed in Figure 9. Unlike the PMF profiles that show a dependence on R_0, the diffusivity profiles of the ions look similar irrespective of the change of R_0 over the entire regions for the cations. It is easily determined that the motion of ions inside the channel is not affected by the increase of R_0 due to confinement, leading to similar diffusivity profiles regardless of R_0. Outside the channel, if the ion diffusivity is governed by local events occurring near an ion, the increase of R_0 will have no effect on the ion diffusivity. Comparison of the flat-bottom half-harmonic restraint with the conventional harmonic restraint method for the position-dependent diffusion coefficients of Na^+ and K^+ indicates that the two methods give rise to the nearly identical results.[14] It is observed that the diffusivity of Cl^- has local minima at both channel entrance regions with $R_0 = 3.5$ Å. This can be explained by the fact that in the channel entrance regions a higher value of R_0 will increase

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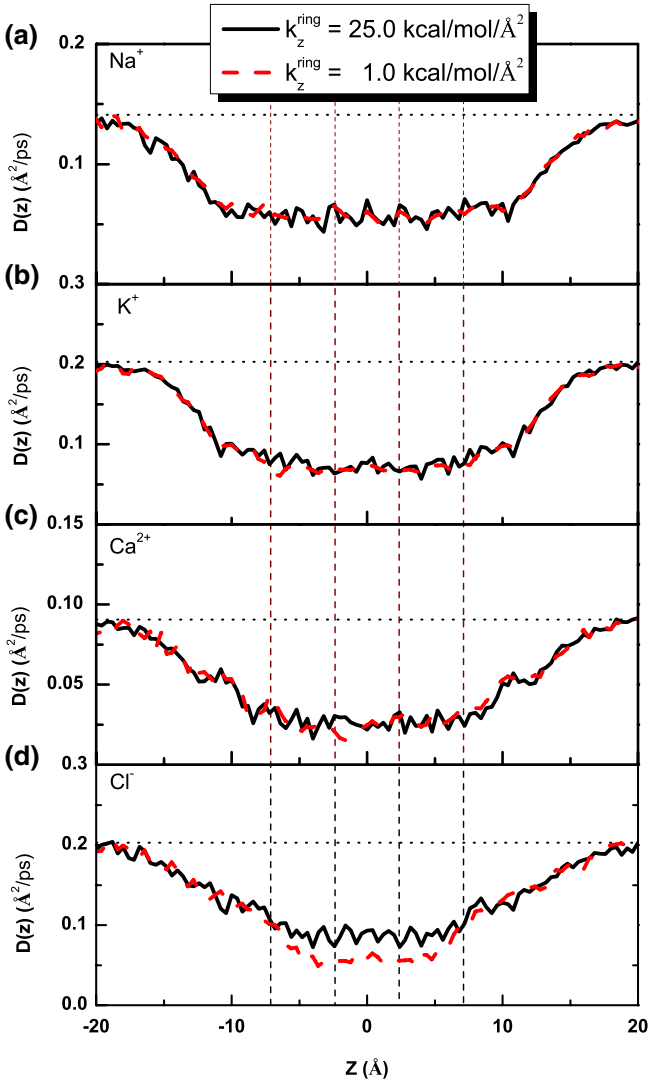


Figure 8. Position-dependent diffusion coefficient profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- representing the dependence of diffusion coefficients on channel rigidity. The channel rigidity was implemented by applying harmonic restraints with a spring constant of k_z^{ring} to the alpha carbons in the backbone of the CPN. For other parameters, $\gamma_L = 4.0$ ps^{-1} and $R_0 = 2.5$ Å were used. The horizontal broken lines indicate the diffusion coefficients of the ions in bulk water with $\gamma_L = 4.0$ ps^{-1}. The vertical broken lines are the same as in Figure 2.

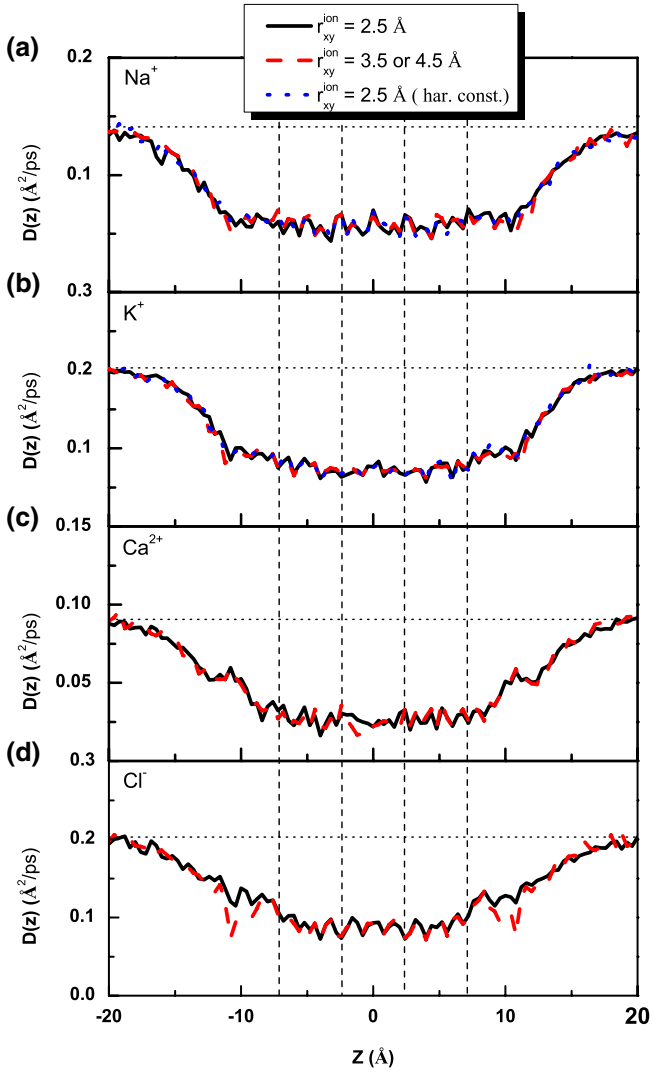


Figure 9. Position-dependent diffusion coefficient profiles of (a) Na^+, (b) K^+, (c) Ca^{2+}, and (d) Cl^- representing the dependence of diffusion coefficients on R_0 in the flat-bottom half-harmonic restraints imposed on the ions in the *xy*-plane. $R_0 = 2.5$ Å was used for all the ions, and $R_0 = 4.5$ Å was used for Na^+, K^+, and Ca^{2+}. In the case of Cl^-, $R_0 = 3.5$ Å was used instead. $\gamma_L = 4.0$ ps^{-1} and $k_z^{ring} = 25.0$ kcal/mol/Å^2 were used. The horizontal broken lines are the same as in Figure 8. The vertical broken lines are the same as in Figure 2.

the repulsive interactions of Cl^- with the dangling carbonyl oxygens from the outermost peptide rings, making it more difficult for Cl^- to enter the channel and thus reducing the diffusivity of the anion.

Concluding Remarks

In this work, the PMF profiles and position-dependent diffusion coefficients associated with transport of Na^+, K^+, Ca^{2+}, and Cl^- through a simple CPN were calculated using MD simulations combined with the ABF method and BI/MC algorithm. The effects of ion valence on the ion selectivity of CPNs were discussed on a basis of the PMF profiles obtained. The dependence of the PMF and diffusivity profiles of the ions on geometric restraints and simulation parameters was also examined.

The PMF profiles for the ions demonstrate the selectivity of the simple CPN to monovalent cations such as Na^+ and K^+. Despite the negative free energy minima in the PMF profiles for Ca^{2+}, those minima are not as deep as those for Na^+ or K^+, and the barriers are higher, so transport of the divalent ion will not be observed. Our analysis shows that among the simulation parameters we examined, the radius parameter that is used in the flat-bottom half-harmonic restraint most significantly affects the energetics

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of the ions through the CPN. However, it also shows that the variation of the PMF profiles due to change of the radius parameter can be predicted using a simple configurational entropy concept. While the damping coefficient in the Langevin thermostat, which is one of dynamics-related simulation parameters, has little influence on the PMF profiles, that parameter greatly affects the position-dependent diffusion coefficients of the ions. However, the effect of the damping coefficient on the diffusion coefficients becomes weaker inside of the CPN than outside. Considering that the channel rigidity has a moderate and minor effect on the PMF profiles and position-dependent diffusion coefficients for ions, we can speculate that the most important factor that affects the position-dependent diffusion coefficients of the cations inside channel is not the damping coefficient or the channel rigidity, but local interactions of ions with water molecules through confinement. This issue will be further investigated in our next study. Finally, our calculations show some dependence of the relative diffusivity of ions on simulation parameters such as the damping coefficient.

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